

Rugby Union Analytics Primer

Rugby Match-Event Data Reference

Rules, Phases, Positions, and Reporting Concepts

Document Scope: Rugby Union (15-a-side) fundamentals for match-event analysis

Primary Audience: Data analytics team preparing a manager briefing

Analytical Focus: Possession, territory, set pieces, breakdowns, discipline, and player roles

Reference Basis: World Rugby Laws of the Game and coaching terminology

Document Type: Practical pre-analysis orientation guide

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1. The game in one minute

Rugby Union is a territorial, possession-contested team sport. Two teams try to move the ball into the opponent's **in-goal** area and score. A player may **run forward** with the ball or **kick forward**, but a hand pass must not travel forward relative to the passer. The defence may tackle the ball carrier; after contact, both teams contest the ball under strict safety and offside rules.

The attacking team repeatedly decides among three broad options:

- **Carry**: take contact and try to retain the ball at a **ruck**;
- **Pass**: move the defence laterally and create space; or
- **Kick**: gain territory, contest a high ball, or force the opponent to exit their own end.

Unlike many invasion sports, possession is not guaranteed after a tackle. The contest at the **breakdown**—the few seconds around a tackle and ruck—often determines both momentum and the next data event.

Know the code before reading the file

Code	Players	Typical format	Analytical warning
Rugby Union	15 (or 7 in Sevens)	2 × 40 minutes for 15s	Rucks, mauls, lineouts and scrums are central.
Rugby League	13	2 × 40 minutes	Six-tackle sets and play-the-balls replace Union's ruck contest.
Rugby Sevens	7	Usually 2 × 7 minutes	Same Union laws with variations; high space and transition rate.

Table 1: Do not merge code-specific metrics without a mapping. This document assumes 15-a-side Rugby Union.

2. Pitch, direction and territory

The rectangular **field of play** is up to 100 m long and 70 m wide. Each end includes an **in-goal** area (up to 22 m). The most useful territorial landmarks in a dataset are the halfway line, the two 10-m lines, the two 22-m lines, the try-lines, and the touchlines. Coordinates can be stored from either team's perspective, so always establish which direction is positive.

Territory is not possession. A team can be dominant in possession but play in its own half; another team can kick effectively and defend from a favourable field position. Report both when possible: for example, possession share, average possession start location, entries into the opposition 22, and points per 22 entry.

3. How a match starts, stops and scores

In the standard adult 15-a-side game, a match has two 40-minute halves, subject to competition regulations. Play begins and restarts after scores with a kick-off. The clock, substitutions, video review procedures and certain technical law applications can vary by competition, so a provider's match clock is the source of truth for timing analysis.

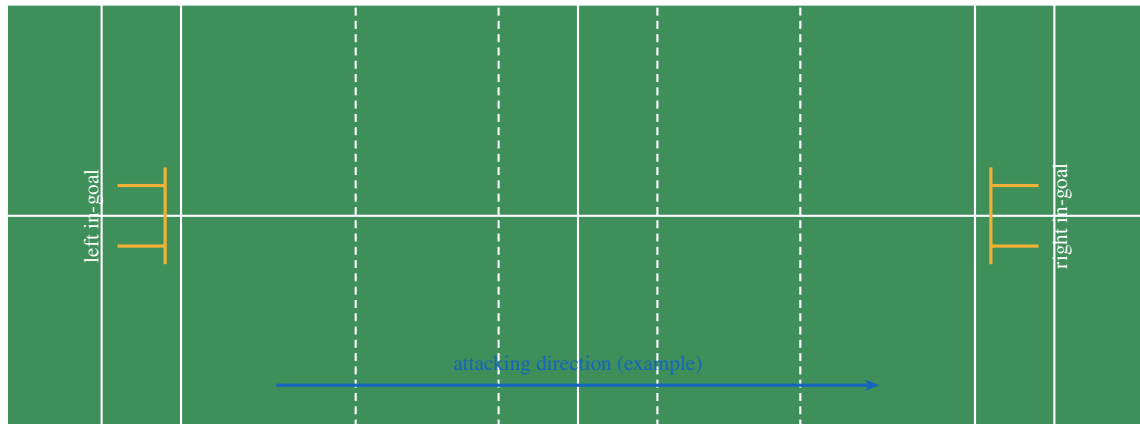
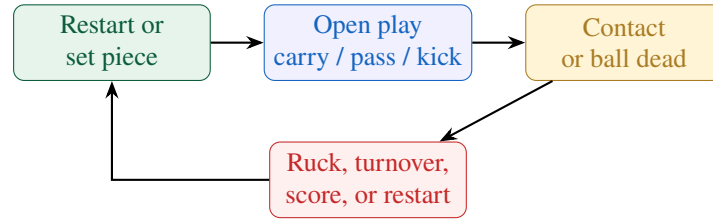


Figure 1: Simplified Rugby Union pitch. A value such as “metres gained” needs a known attacking direction; otherwise the sign may be reversed.

Score	Points	How it happens	Useful data interpretation
Try	5	Ball is grounded with downward pressure in opponents’ in-goal.	The main attacking outcome; separate tries from attempted tries if available.
Conversion	2	Place-kick or drop-kick after a try.	Usually linked to the try’s lateral location.
Penalty goal	3	Successful kick at goal after a penalty.	Often reflects field position and decision-making, not just attacking quality.
Drop goal	3	Dropped and kicked through the posts during open play.	Rare, often late-game or in close territory battles.

Table 2: Scoring values in Rugby Union. A penalty can instead be tapped, kicked to touch, or kicked for territory.

The event loop



An **event log** breaks this loop into records (for example: kick-off, carry, tackle, ruck, pass, kick, penalty, lineout, score). A **possession** or **sequence** groups consecutive events by one team until it scores, kicks the ball away, loses the ball, or play stops. Provider definitions matter: ask how it defines a possession, ruck, tackle break, dominant tackle, and turnover.

4. Fifteen positions and their jobs

Numbers are traditional shirt numbers, not rankings. Players can have wider skill sets than their labels suggest, but position is a useful first grouping for a CSV.

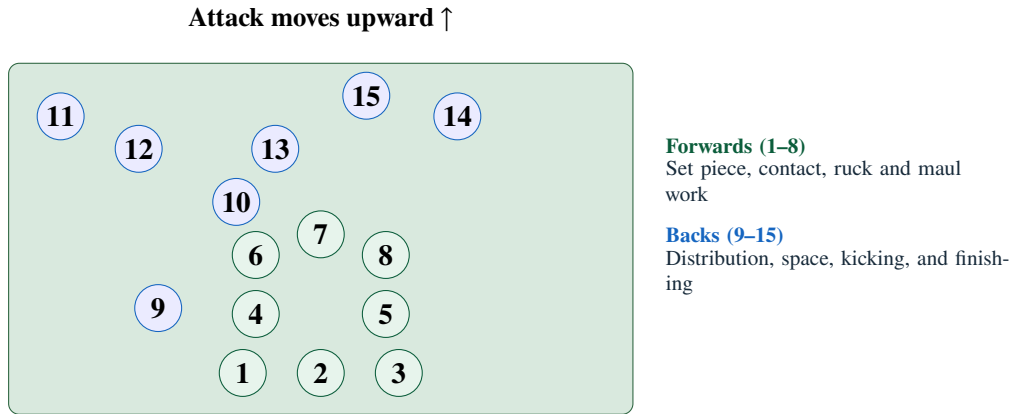


Figure 2: A conventional attacking shape, not a fixed map. Rugby is fluid after the first phase.

No.	Position	Unit	Typical contribution
1, 3	Loosehead / tighthead prop	Front row	Scrum stability; lifting and clearing at contact.
2	Hooker	Front row	Throws into lineouts; hooks ball in scrums; heavy contact work.
4, 5	Locks (second row)	Tight five	Lineout targets and lifters; scrum power; high work rate.
6, 7	Flankers	Back row	Tackle, contest and support; often different blindside/open-side roles.
8	Number eight	Back row	Links scrum to attack; carries and controls ball at scrum base.
9	Scrum-half	Half-back	Retrieves and distributes from rucks, scrums and lineouts.
10	Fly-half	Half-back	Tactical distributor/kicker; often directs the attack.
11, 14	Wings	Back three	Width, finishing, aerial contests and kick chase.
12, 13	Centres	Midfield	Carrying, passing, defensive alignment and line breaks.
15	Fullback	Back three	Back-field coverage, high balls, counter-attack and kicking.

Table 3: Position groups are more robust for a first dashboard than player comparisons across dissimilar roles.

5. The key laws behind most event labels

Passing, kicking and the forward-pass rule

A thrown or handed pass must not go forward. The ball may travel forward after a kick, and a player may run forward. A ball that travels forward from a player unintentionally (a **knock-on**) normally stops play and often results in a scrum to the opposition. Forward-pass and knock-on records can be useful error measures, but they are not interchangeable.

Tackle, ruck and maul

Situation	Plain-English definition	What an analyst should look for
Tackle	Ball carrier is held and brought to ground by an opponent. The tackle law obligations then apply.	Tackle location, tackle completion/miss, metres after contact, and speed of recycle.
Ruck	At least one player from each team is on their feet, in contact, close around a ball on the ground.	Retention, turnover conceded/won, ruck speed, number of attackers/defenders, and penalty outcome.
Maul	A ball carrier and at least one player from each team are bound together and remain on their feet.	Start location, metres made, collapse/penalty, exit method, and try probability (especially near goal).

Table 4: A tackle may lead to a ruck, but not every tackle has a ruck. Keep those denominators separate.

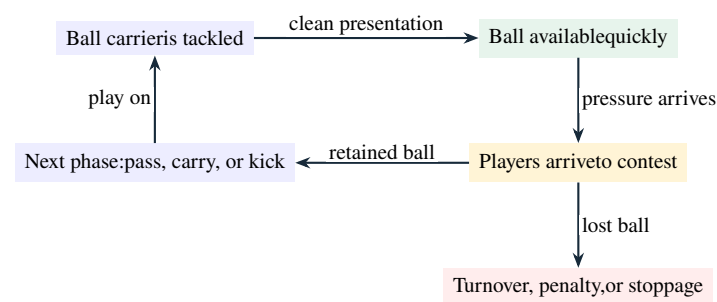


Figure 3: Simplified post-tackle decision tree. The quality and timing of the next arriving players changes the outcome.

Offside, advantage and penalties

At a ruck, maul, scrum or lineout, players must respect an **offside line**; they cannot interfere from an offside position. In open play, a player ahead of a team-mate who last played the ball can also be offside until put onside. Penalties result from many infringements (including offside, illegal tackle, not releasing, or collapsing a set piece). The referee may play **advantage** instead of immediately stopping play when the non-offending team has a real opportunity to benefit.

For analysis, distinguish **penalties conceded** (discipline cost) from **penalties won** (which may reflect opposition indiscipline or pressure). Also preserve whether a penalty was taken as a goal attempt, kick to touch, scrum, or tap: those choices have different intent.

6. Set pieces: organised chances to win possession

Scrum

A scrum restarts play after certain stoppages, commonly a knock-on or forward pass. Eight forwards from each team bind in three rows; the scrum-half feeds the ball into the middle. The team that did not cause the stoppage usually receives the feed. It is a contest, but the fed team normally has the possession advantage.

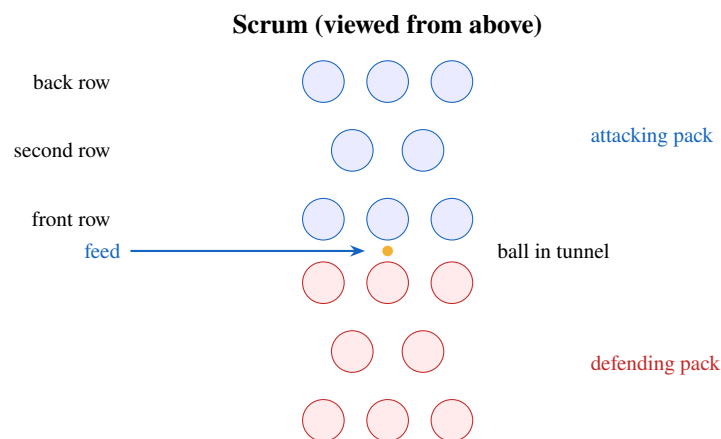


Figure 4: Eight forwards per side form a scrum. In provider data, separate scrum wins/losses, penalties, free-kicks and resets.

Lineout

When the ball goes into **touch** (out over a sideline), play usually restarts with a lineout. Forwards form two lines near the touchline; the throwing team tries to hit a nominated jumper, often lifted by team-mates. A lineout is both a possession event and a territory event because it occurs where the ball entered touch.

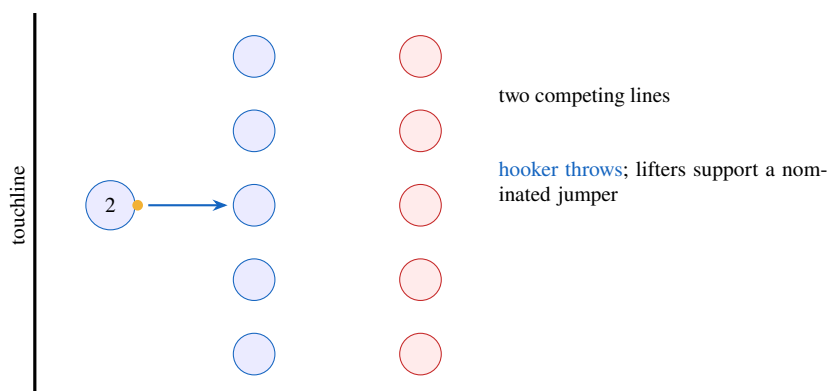


Figure 5: Simplified lineout. Spacing is deliberately exaggerated so the two competing lines and throw are clear.

7. A practical analytics translation layer

Typical fields and questions

Event providers use different schemas; some log every pass while others do not. Map the real columns before calculating a KPI.

Starter KPIs—with explicit denominators

- **Possession retention rate** = $\frac{\text{possessions retained after a contestable event}}{\text{such possessions}}$. State precisely whether the denominator is rucks, lineouts, all possessions, or another definition.
- **Tackle completion** = $\frac{\text{completed tackles}}{\text{completed + missed tackles}}$. Check whether assisted tackles are counted once or per player.
- **Gain-line success** = $\frac{\text{carries crossing the gain line}}{\text{carries}}$. Do not call it “line breaks”; a line break is usually a much rarer event.
- **Lineout success** = $\frac{\text{own lineouts retained}}{\text{own lineouts thrown}}$; report by throwing team, not all lineouts.
- **Scrum success** = $\frac{\text{own feeds retained/awarded}}{\text{own feeds}}$. Depending on the provider, include outcomes such as penalties separately rather than calling all of them wins.

Likely field / concept	Meaning	Question it supports
event_type, subtype	What happened and its classification.	Which actions create scores, errors or territory?
team, player, position	Who performed the event.	Who carries workload; which unit is driving outcomes?
period, clock	When it happened.	Does discipline, pace or effectiveness change by half/quarter?
x, y, end_x, end_y	Start/end location.	Where are attacks initiated, ended, or lost?
possession_id, phase	Sequence and phase within it.	Does attack improve or decay across phases?
outcome, turnover, penalty	Result of the action.	What drives retained ball and avoidable concessions?
metres, gainline	Progression indicator, if provided.	Who/what gets over the gain line or makes post-contact metres?

- **Penalty rate** = $\frac{\text{penalties conceded}}{\text{minutes played}} \times 80$, or per possession/ruck when modelling discipline. Declare the exposure.
- **22-entry efficiency.** $\frac{\text{points or tries from possessions entering opposition 22}}{\text{opposition 22 entries}}$ Decide and state a time/possession attribution window.

A manager-ready storyline

Build the report in this order: **scoreboard outcome** → **territory and possession** → **set piece and breakdown** → **discipline/errors** → **players and tactical examples**. This prevents a high-volume statistic (for example, carries) being presented as performance without showing whether it produced field position, retained possession, or points.

Example insight wording. “Team A had fewer possessions but began 41% of them in the opposition half. It converted 5 of 7 entries into the 22 into points, while Team B lost 4 of 12 rucks inside its own half. The decisive pattern was therefore territory plus breakdown retention, not raw possession.” Replace all figures with values calculated from the supplied data and document the provider definitions.

8. Data traps worth checking first

1. **Team perspective:** Some feeds invert x/y by half; normalise all attacks toward one end before plotting heat maps or metres gained.
2. **Clock versus elapsed time:** A display clock may run down, while an event clock may count up and include stoppage time. Sort with period as well as clock.
3. **Duplicate or linked events:** A try may be represented by a grounding event, score event and conversion event. Link rather than double-count.
4. **Outcome vocabulary:** “Won”, “retained”, “complete” and “success” can mean different things by event type. Inspect distinct values and data documentation.
5. **Small samples:** One scrum penalty or one intercept can dominate a single-match percentage. Always display the numerator and denominator alongside the rate.
6. **Causality:** A player’s tackle count often reflects where the opposition attacked, not necessarily poor or excellent defending. Combine volume, success, field zone and video/context where available.

9. Glossary

Advantage

Referee allows play to continue after an infringement if the non-offending team may benefit.

Breakdown

The contest immediately around a tackle, including the ensuing ruck where formed.

Carry

A run with the ball; may finish in contact, a pass, a kick or a score.

Gain line

An analytical reference line, normally where the phase began; it is not a painted field line.

In-goal

Area beyond the try-line where a try can be scored by grounding the ball.

Knock-on

Loss of possession when the ball goes forward from a player's hand/arm, subject to law detail.

Line break

A clean penetration through the main defensive line; not

simply a successful carry.

Offload

A pass made while in contact, before the ball carrier is brought to ground.

Phase

A segment of attack, often the play after a ruck, maul, scrum or lineout.

Ruck

On-feet contest over a ball on the ground involving at least one player from each team.

Set piece

Structured restart: principally scrum or lineout.

Touch

The sideline/out-of-bounds area; ball in touch normally leads to a lineout.

Turnover

A change of possession, through a steal, error, interception, failed set piece, penalty, etc. Define the subtype.

10. Authoritative references

Rules evolve. For any production report, verify the competition's current regulations and the provider's definitions against the official laws.

- World Rugby, *Laws of the Game* (current law text and law-by-number pages).
- World Rugby, *Teams and players* (position overview).
- World Rugby, *Ruck and maul functional roles* (breakdown concepts).